FEDERATION UNIVERSITY AUSTRALIA · ITECH3208 IT CAPSTONE PROJECT 1

Sprint 3 *Report.*

Pedagogy Evaluation Platform — a web-based self-assessment tool implementing Firmin's (2020) Techno-Pedagogical Practice framework. Sprint 3 delivered optional demographic collection, an expanded administrator dashboard, and survey completion validation, as specified in the client planning meeting.

UNIT
ITECH3208 IT Capstone Project 1

SPRINT
Sprint 3

STATUS
Implementation complete

DATE
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PROJECT TEAM

Mushfiqur Rahman	Lead Developer
Shafiullah Anik	Scrum Master
Albin Roy	Product Owner
Merajun Nabi	Tester

CONTENTS

01 **Summary**
02 **Sprint objectives**
03 **Demographic collection**
04 **Administrator dashboard**
05 **Validation**
06 **Quality assurance**
07 **Testing**
08 **Outstanding items**
09 **Appendix**

01

Summary

Sprint 3 addressed three requirement areas raised by the client representative in the sprint planning meeting: optional demographic data collection, a substantially expanded administrator dashboard, and stricter validation of survey completion. All three were implemented, reviewed, and verified against the acceptance criteria set in that meeting.

Implementation is complete. A formal test script covering every acceptance criterion has been prepared for the tester, and the build is ready to enter the testing phase. Section 08 records the items scheduled to follow: the database migration, deployment, and the one verification step that depends on the migration.

REQUIREMENT AREA	OUTCOME	STATUS
Optional demographic collection	Consent question presented on survey completion, followed by a dedicated page capturing age group, gender and academic level, linked to the same survey response.	DELIVERED
Administrator dashboard	Eight summary statistics, a numerical breakdown of the four framework categories, and three demographic distribution charts.	DELIVERED
Validation and error handling	Per-question validation, an answer review screen, and a single completion gate that prevents partial results from being scored or stored.	DELIVERED
Test documentation	A written test script covering the five verification items specified by the client representative, plus dashboard and regression coverage.	DELIVERED
Database migration	Schema definition for the new demographics table, written and reviewed, scheduled for execution ahead of release.	SCHEDULED

02

Sprint objectives

As specified in the Sprint 3 planning meeting.

REF	REQUIREMENT AS STATED
R1	After the survey is completed, ask whether the participant is willing to provide demographic information. If they agree, present an additional page collecting those details, linked to the same survey response. Initial fields: age group, gender (with an option to specify), and academic level A–E.
R2	Focus the sprint on the administrator dashboard, displaying summary statistics: number of participants, number who started the survey, number who completed it, number who created accounts, a summary of the TP / PD / TA / TPP results, and basic demographic summaries, preferably as bar charts.
R3	Improve survey validation: participants cannot submit unless all questions are answered, an error message is shown if any question is incomplete, and results are saved only when the survey is fully complete.
R4	Verify by testing: incomplete survey submission, error message behaviour, completion button functionality, correct score calculation, and the demographic page flow and data saving.

03

Demographic collection — R1

Participant flow

On completing the survey the participant is asked *"Are you happy to provide demographic information?"* Selecting **Yes** opens a dedicated page; selecting **No thanks** proceeds directly to their results. Both paths reach the results page in a single action, so the optional step never obstructs the participant.

Fields collected

FIELD	OPTIONS PRESENTED
Age group	Under 25 · 25–34 · 35–44 · 45–54 · 55–64 · 65 or over · Prefer not to say
Gender	Male · Female · Other, with a free-text description limited to 60 characters · Prefer not to say
Academic level	Level A (Associate Lecturer) · Level B (Lecturer) · Level C (Senior Lecturer) · Level D (Associate Professor) · Level E (Professor) · Other · Prefer not to say

Data model

- A new `demographics` table stores one record per survey session, joined to the score record by session identifier. This satisfies the requirement that the information be linked to the same survey response.
- Declines are recorded explicitly. Without this the opt-in rate could not be reported, as a participant who declined would be indistinguishable from one never asked.
- Records are write-once. No update or delete permission is granted, so possession of a session identifier does not permit a record to be altered.
- Permitted values are enforced by database constraints as well as by the interface, so malformed data cannot be introduced by bypassing the form.

ETHICAL CONSIDERATIONS

Every question offers a *"Prefer not to say"* option, so requiring an explicit selection never compels disclosure. Responses are attached to the assessment record rather than to a participant's name, and participants using the anonymous access route may contribute demographic data without creating an account.

04

Administrator dashboard — R2

Summary statistics

Eight figures are presented in two groups. Definitions are stated explicitly because several of these counts could reasonably be interpreted in more than one way.

STATISTIC	DEFINITION APPLIED
Participants	All non-administrator individuals known to the platform, whether registered or anonymous
Started survey	Participants with at least one assessment session recorded
Completed survey	Participants with at least one session marked complete
Accounts created	Participants who registered an account, as distinct from anonymous access
Completion rate	Completed divided by started
In progress	Started but not yet completed
Average score	Mean overall score across completed assessments
Demographics	Count who provided details, with the declined count and resulting opt-in rate

Framework category results

The existing category bar chart is now accompanied by a numerical table reporting the count of responses, mean, median, lowest and highest value for each of the four framework categories — Teaching Practice, Pedagogical Development, Technology Adoption and Techno-Pedagogical Practice — together with an overall composite. Both the chart and the table are computed from the same set of completed sessions, so the two presentations cannot disagree.

Demographic summaries

Three bar charts present the distribution of age group, gender and academic level across participants who opted in. Categories with no responses are omitted so that small samples remain legible. Demographic data is also surfaced in the participant table, in individual participant records, and as four additional columns in the exported data file for subsequent analysis.

DEFECT IDENTIFIED DURING IMPLEMENTATION

Assessment sessions that could not be matched to a participant record were being excluded from the dashboard entirely. These are now reported as unattached sessions. Had this gone uncorrected, the participant count required by R2 would have under-reported the true total.

05

Validation and error handling — R3

Two points of validation

- **At each question.** Attempting to advance without selecting a response displays an explicit message — *"Please select a response before continuing — every question is required."* — in place of the previous behaviour, which silently disabled the button and gave the participant no explanation. The message clears as soon as a response is selected.
- **At submission.** The final question leads to a review screen listing all twenty responses grouped by framework category. Attempting to submit while incomplete is refused, and the resulting message identifies the specific unanswered question numbers and links directly to the first of them.

Results saved only on completion

A single point in the code governs this. No score is calculated, stored, or transmitted until all twenty questions hold a valid response. This was confirmed by attempting an incomplete submission and verifying that neither the local record nor the database contained any result.

DESIGN DECISION

Because each question must be answered before the participant can advance, the submission step could not otherwise be reached in an incomplete state, making R3 impossible to demonstrate. A *Review answers* control was therefore added to the progress bar, allowing the review screen to be opened at any point. This both makes the rule testable and lets participants check their progress before submitting.

06

Quality assurance

Four defects were identified during a structured review of the new code and resolved before the build was passed to testing.

1 Save failures were reported to the participant as successes

`Data integrity · js/assessment.js`

The database client reports failures through a returned value rather than by raising an exception, and that value was not being examined. A rejected write — whether from a permissions rule, a constraint violation, or loss of connectivity — passed unnoticed, and the participant was informed their results had been saved. Both the score update and the response insertion are now checked, and the completion screen reports the true outcome in either case.

2 Navigating away could cancel the save in progress

`Data loss · js/questionnaire.js`

The save was performed in the background while the completion screen already offered controls that navigate to another page, so a prompt selection could terminate the request before it completed. Those controls are now held, with a visible progress indicator, until the write has settled, subject to a six-second limit so that a participant is never detained by an unresponsive connection.

3 Validation messages could attach to the wrong question

`Correctness · js/questionnaire.js`

Two question panels exist briefly during the transition animation. An error message could therefore be attached to the panel leaving the screen rather than the one presented to the participant. All references are now bound to the active panel.

4 Returning from the review screen moved the participant to the final question

`Usability · js/questionnaire.js`

Opening the review screen partway through the survey and returning placed the participant at question twenty rather than at the question they had left. The original position is now preserved.

07

Testing and verification — R4

Developer verification performed

Conducted in a browser against the local build prior to handover for testing.

AREA VERIFIED	RESULT
Incomplete submission	Refused, with the unanswered questions identified; no score and no database record written
Error messages	Displayed on the correct question, cleared on selection, and did not permit advancing
Score calculation	Verified against manual calculation: responses of 5, 4, 2 and 1 by category produced 100, 80, 40 and 20 respectively, with an overall score of 60, matching the results page
Demographic flow	Accept, decline and skip paths, the free-text gender option, and the safeguard preventing direct access without a completed survey
Dashboard statistics	All eight counts checked by hand against a controlled data set; the category composite matched the average score reported
Degraded operation	Dashboard loads completely and reports the reason when the demographics table is absent, rather than failing
Keyboard interaction	Number, arrow and enter keys behave correctly, including refusing to advance past an unanswered question
Responsive layout	Review screen, demographic form and dashboard examined at 375 px width
Runtime errors	None observed in the browser console

Formal testing

A written test script, `SPRINT3_TEST_PLAN.md`, has been prepared for the tester. Sections 1 to 5 correspond to the five verification items specified under R4; sections 6 and 7 extend coverage to the dashboard and to regression checks on existing functionality. Each step states the action and the expected result, with explicit pass criteria.

> **LIMITATION OF DEVELOPER VERIFICATION**
>
> The development environment used for this verification had no network route to the hosted database, so the round trip to the live database could not be exercised. Client-side behaviour is verified; confirming that a demographic record is written to the database is covered by step 5.1.9 of the test script and is carried out during the testing phase.

08

Outstanding items

Sequenced work remaining at the close of Sprint 3.

#	ITEM	DETAIL
1	**Database migration**	The demographics table is defined in `sprint3-demographics.sql`, written and reviewed. It is additive: it creates one table with its access rules and alters no existing table. It is executed against the project database before release.
2	**Testing phase**	The build enters testing against `SPRINT3_TEST_PLAN.md`. Step 5.1.9, confirming a demographic record reaches the database, follows item 1.
3	**Deployment**	The build is complete and verified locally but has not been released. Release follows test sign-off, in line with the project's practice of not publishing directly to the live site.
4	**Email recipient**	Results notifications are currently directed to a development address to avoid sending test data to the client. This is changed to the client's address before participants are recruited. Carried forward from an earlier sprint.

Considerations for the next sprint

- Cross-tabulation of framework scores against demographic groups, which is the analysis the demographic data was collected to support.
- Extending the demographic summaries into the exportable report for the client.
- Ethics consent recording, identified in earlier planning and not yet scheduled.

09

Appendix — files added and modified

FILE	CHANGE	PURPOSE
sprint3-demographics.sql	Added	Demographics table, value constraints and access rules
demographics.html	Added	Demographic collection page
js/demographics.js	Added	Form rendering, validation and saving
js/demographic-options.js	Added	Shared definition of the permitted values
css/demographics.css	Added	Demographic page presentation
SPRINT3_TEST_PLAN.md	Added	Test script for the sprint acceptance criteria
js/questionnaire.js	Modified	Validation, review screen and consent step
js/assessment.js	Modified	Save-outcome reporting and demographic persistence
js/admin.js	Modified	Statistics, category summary, demographic charts, export
admin.html	Modified	Dashboard structure and new panels
questionnaire.html	Modified	Review control in the progress bar
css/questionnaire.css · css/admin.css	Modified	Review screen, consent panel and dashboard layout